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Course: Calculus Beta

Assignment: MyLab Math - Uge 7

A study on body types gave the following results: 46% were short; 23% were short and overweight; and 22% were tall and not overweight. Find the probabilities that a person is the following.

- a. Overweight
- b. Short, but not overweight
- c. Tall and overweight

a. To find the probability that a person is overweight, $P(\text{overweight})$, add the probability that a person is short and overweight and the probability that a person is tall and overweight.

To find the probability that a person is tall and overweight, first find the probability that a person is tall.

$$\begin{aligned} 1 - P(\text{short}) &= 1 - 0.46 \\ &= 0.54 \end{aligned}$$

Next find the probability that a person is tall and overweight.

$$\begin{aligned} P(\text{tall and overweight}) &= P(\text{tall}) - P(\text{tall and not overweight}) \\ &= 0.54 - 0.22 \\ &= 0.32 \end{aligned}$$

Now find the probability that a person is overweight.

$$\begin{aligned} P(\text{overweight}) &= P(\text{short and overweight}) + P(\text{tall and overweight}) \\ &= 0.23 + 0.32 \\ &= 0.55 \end{aligned}$$

b. To find the probability that a person is short, but not overweight, $P(\text{short, but not overweight})$, subtract the probability that a person is short and overweight from the probability that a person is short.

Find the probability that a person is short, but not overweight.

$$\begin{aligned} P(\text{short, but not overweight}) &= P(\text{short}) - P(\text{short and overweight}) \\ &= 0.46 - 0.23 \\ &= 0.23 \end{aligned}$$

c. Find the probability that a person is tall and overweight. Recall that the probability that a person is tall and overweight was found in part (a).

$$P(\text{tall and overweight}) = 0.32$$